

CoE 1199 Senior Project

ChessConnect

Final Report

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Introduction

Chess is a game that is traditionally played over-the-board, but as humanity has transitioned into the digital age the game has undergone many transformations that have affected how players experience it. The most apparent of these changes has been the rise of online chess. With the popularity of personal computers and phones, more and more people are turning to digital chess and abandoning the traditional format.

As a result, the two formats have evolved in ways that have made them increasingly distinct from one another. Online chess is often more convenient for users, providing them with the ease of finding a game against an opponent of similar skill within seconds. Despite this convenience, the format also comes with drawbacks that can impact one's performance. For example, many players find it easy to get distracted while playing online chess due to being plagued by notifications, pop-ups, and advertisements which can easily break their focus. This shorter attention span is also highlighted by the fact that online chess players prefer games with shorter time controls. Online, fast-paced games do not effectively give players time to fully calculate long sequences of moves and figure out a plan — a skill that is necessary for longer time controls, which are typically used in over-the-board games. These differences in pace and environment illustrate some of the broader distinctions between online and traditional chess; however, even the core skillset required to play differs between the formats.

In traditional chess, players study and visualize a physical, three-dimensional board. In contrast, online chess requires visualization of a two-dimensional board shown on a screen. The pattern recognition necessary to visualize the board for each format is very different, as the physicality of over-the-board games requires a level of spatial awareness that is not present when playing online games. As such, the two formats require distinct ways of thinking and recognizing

patterns. This becomes apparent when online users play an over-the-board game, wherein they may struggle with visualizing potential moves and vulnerabilities to their pieces. Many online chess platforms even give users the ability to draw arrows on their board, removing a key aspect of the visualization required when playing an over-the-board game. As a result, these differences between the two formats often make it difficult for one to transition from online to traditional chess without facing challenges stemming from both digital and physical limitations.

The solution to this disparity is to create an electronic chess set that can be used to translate virtual chess games into the physical realm and vice versa. Combining the aspects of both online and traditional chess requires a specific balance between electronic assistance and additional complexity to the traditional chess experience. This must be achieved while ensuring the product remains convenient, easy to use, and affordable. It is this balance that ChessConnect strives to achieve.

Solution

ChessConnect is an electronic chessboard designed to bring a new and innovative experience to chess players. Using the board, a chess game can be played against an online opponent while the ChessConnect user plays on a physical board. When the user makes a move, the board will automatically relay it to the opponent's online game. Similarly, when the online opponent makes a move, the pieces on the physical board will automatically move to their updated positions. In doing this, ChessConnect allows for the convenience of online chess games while allowing users to form good habits that are built from playing physical chess games.

As initially proposed, ChessConnect was intended to connect to an online chess service using an Arduino Mega microcontroller along with a WiFi module. Once the board has an established connection, the board would then connect to Chess.com to allow the user to play

against online opponents while at a physical board. Move detection for the user would be done using reed switches placed on each chess square on the underside of the board. These reed switches would interface with magnets internal to each chess piece so that the state of the board can be tracked in real time by scanning these switches. Once it becomes the opponent's turn, the board would connect with Chess.com and wait for a move to be sent by the opponent. Once a move is sent, an internal gantry with an attached electromagnet would position itself under the piece to be moved. It would then move this piece by turning the electromagnet on and moving to the piece's destination square. In the case of taking pieces or knight movement, pieces would be sized to allow for movement on the border of squares without interfering with other pieces on the board.

However, now that the device has been built, changes have been made to the preliminary design of the device. The device functions nearly the same as proposed, with little to no functional changes being made to the core concept of the project. However, a change needed to be made to the server on which the ChessConnect user will play games. Originally, the device was to connect to Chess.com, but during testing, it was determined that the Chess.com API was very closed off to the public. To gain access to the API, multiple steps of documentation were needed and resources pertaining to the use of the API were sparse. As a result, Lichess.org, another large online chess service with an open access API, was determined to be a more suitable server to be used for the device. This API also came with extensive documentation and examples which made implementation much less time consuming. Along with the server change, hardware modifications were also made to the device to make online capabilities possible. During testing, the project group determined that the Arduino Mega and WiFi module are not capable of interfacing with the Lichess server. While these components can successfully connect to WiFi

networks and connect to multiple online servers, they are incapable of connecting to Lichess due to the type of connection encryption used with their servers. To connect to Lichess, a root certificate must be stored in the microcontroller, and the Arduino Mega lacks the processing power and memory to handle this encryption process. To resolve this, the Arduino Mega and WiFi module was returned and substituted with an Arduino UNO R4 WiFi. This microcontroller was chosen due to its built-in ESP32 chip which is capable of interfacing with encrypted servers like Lichess. These changes were crucial to the successful implementation of online connectivity onto the physical board.

Results

The completed ChessConnect system successfully fulfills its intended functionality of bridging the gap between online and traditional chess formats. Once powered and connected to WiFi and a Lichess.org account, the board enables users to start a new game directly from the onboard interface. Once the match begins, the board operates exactly as specified during the design phase. When it is the opponent's move, the internal gantry will move under the moving piece's square, take it to its new square without interfering with uninvolved pieces on the board, and position it on the destination square. When it is the user's turn, the board will continuously scan the reed switches and update the board state until a change is made. If the user does not wish to make a move and wants to view options, they can browse a brief options menu allowing them to resign the game or adjust pieces manually if many pieces have been knocked off their space. Otherwise, once a piece is picked up, it will be determined whether the user is taking a piece or not along with allowing them to return their piece back to its original square and make a different move if they wish. Once a move is made, it is sent to the Lichess server and reflected in

the ongoing online game. This bidirectional synchronization enables a seamless playing experience and gives users the benefits of both online and traditional chess formats.

In addition to the core features, ChessConnect accounts for special cases that can happen during a chess game. All unique chess moves are accounted for when making an automatic move on the board or a user move which ensures that any type of move can be played. Furthermore, if the user makes an illegal move, the board includes safeguards to maintain the gameplay flow and allow the user to remedy their mistake. If an illegal move is made, the user is informed of the invalid move via the LCD screen and prompted to put the piece(s) back. Once they have done this, they can return to the gameplay and make another move as if the illegal move had never happened.

Conclusion and Recommendations

ChessConnect has reinforced many ideas and techniques learned throughout the curriculum at the University of Pittsburgh at Johnstown. These include many different hardware and circuit design skills, microcontroller coding and usage, and general software implementation and practices. Along with concepts, many new skills were developed by the group members to complete the project. These skills include soldering, API integration/interfacing, and automation controls. The team also developed crucial testing skills which were required to take ChessConnect from an interesting concept to a functioning prototype. For example, extensive testing needed to be conducted surrounding the magnetic interactions that allowed the device to work. The project team had to repeatedly test and adjust the various implementations of the magnetic switches, electromagnet, and internal piece magnets to strike a balance that was not only functional but also consistent. These tests also expanded the team's knowledge of various magnetic technologies which provided valuable insight into future changes which could

potentially improve the device's functionality. Overall, ChessConnect proved not only to be a successful project but an important learning experience that highlighted the importance of adaptability, hands-on development, and continuous learning.

Looking ahead, there are some future improvements that can be made to ChessConnect to enhance the gameplay experience for the user. General build quality improvements would greatly help not only with the overall appearance of the device but also improve functionality in nearly all areas. Better reed switches and more suitable material to construct the chassis would allow for more consistent piece detection, and better motors, perhaps with dampening, could also be used in the future to reduce the vibration and sound produced during gantry movement. Another future consideration is compatibility with different chess servers and game modes. In the future, the project group would like to interface with other online servers like Chess.com to allow users to have the most freedom when choosing how to play. Also, more game modes could be implemented in the future such as game modes with unique starting positions or the development of a ChessConnect specific game mode.

Hardware Appendices

Description of Components

Listed below are descriptions of the core hardware components of ChessConnect and the function they serve to the overall device. The interaction between these components is expanded upon in the following section – the block diagram.

Arduino UNO R4 WiFi - The Arduino microcontroller enables connection to Lichess and acquires data from the various inputs to facilitate the user interface (UI) as well as interpret the user's chess moves which are then sent to the online game. Similarly, it converts the opponent's move from the online game into the corresponding sequence of motor and electromagnet controls that update the piece positions.

LCD Display - The LCD Display facilitates all direct communication with the user, allowing them to traverse through the game mode selection, check their move before sending, display errors or illegal moves, and view options during a game.

Speaker - The internal speaker is a small, two wire piezo buzzer used solely for making simple beeps when a piece is not detected on its destination square after an automatic move.

12V DC Power Supply - The power supply is used to convert 120V AC power from a common wall outlet into the 12V DC power needed for various components.

Push Button (x2) - The push buttons are located on the black and white sides of the LCD display. They are used by the user to navigate various options listed on the LCD display and to send a move.

BJT – TIP 120 - The TIP-120 bipolar junction transistor (BJT) functions as a control switch for the electromagnet. The higher current and voltage demands of the electromagnet necessitates that it is powered from the 12V supply directly instead of from the Arduino pins. By connecting the base pin of the BJT to the microcontroller and routing the supply power to the electromagnet through the collector and emitter pins, the Arduino can control when the current to the electromagnet flows.

Electromagnet - The electromagnet is used on the internal gantry to control piece movement. When the opponent's move is received, the magnet will be positioned beneath the piece, turn on, and hold the piece as it is moved to its destination.

Stepper Motor Driver (x2) - The stepper motor drivers are used to operate the motors for the XY rail system. These motor drivers take instructions from the microcontroller and convert them into movements that the motor can perform.

Stepper Motor (x2) - The motors are used to facilitate the movement of the electromagnet mounted on the XY rail system and allows for piece movement.

Reed Switch (x64) - The Reed switches are used to track the positions of the pieces during a game. These switches are located under every square of the board, allowing them to interact with the internal magnets of each piece. By tracking the changes in which switches are closed or open, the microcontroller can detect how the user moves their pieces.

MUX (x4) - The 16-channel multiplexers are used to reduce the number of pins necessary to connect the 64 reed switches to the microcontroller.

Micro Switch (x2) - The micro switches are mounted onto the gantry and are used as limit switches for calibration. They are placed in specific positions such that the rail system can move the electromagnet to the furthest corner of the rail system. This allows the electromagnet to have a known position to move from

Block Diagram

The block diagram shown below in Figure 1 details the connections between the hardware components of ChessConnect.

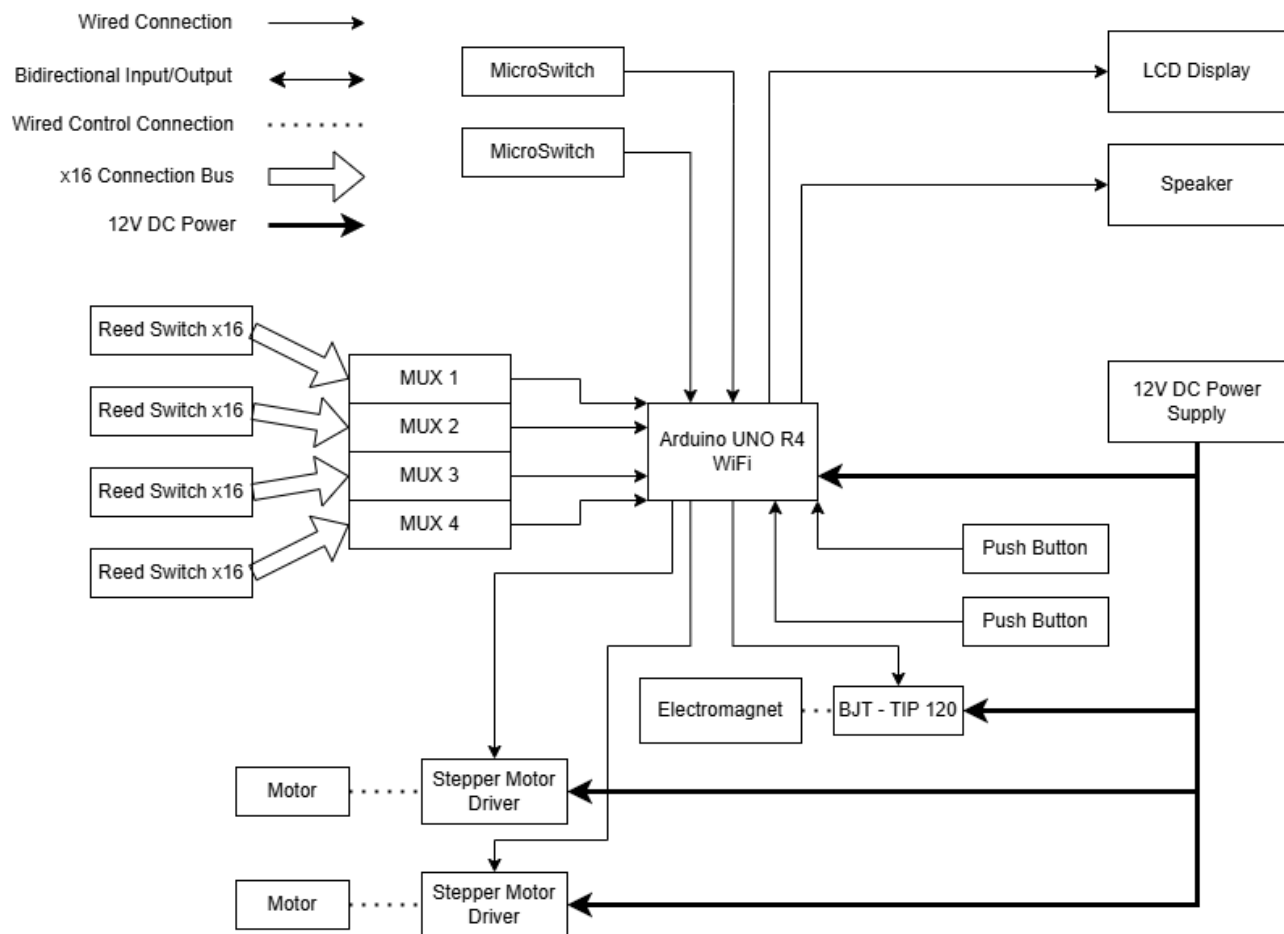


Fig. 1. Block Diagram

Circuit Diagram

The circuit diagram shown below in Figure 2 outlines the connections between each hardware component described above.

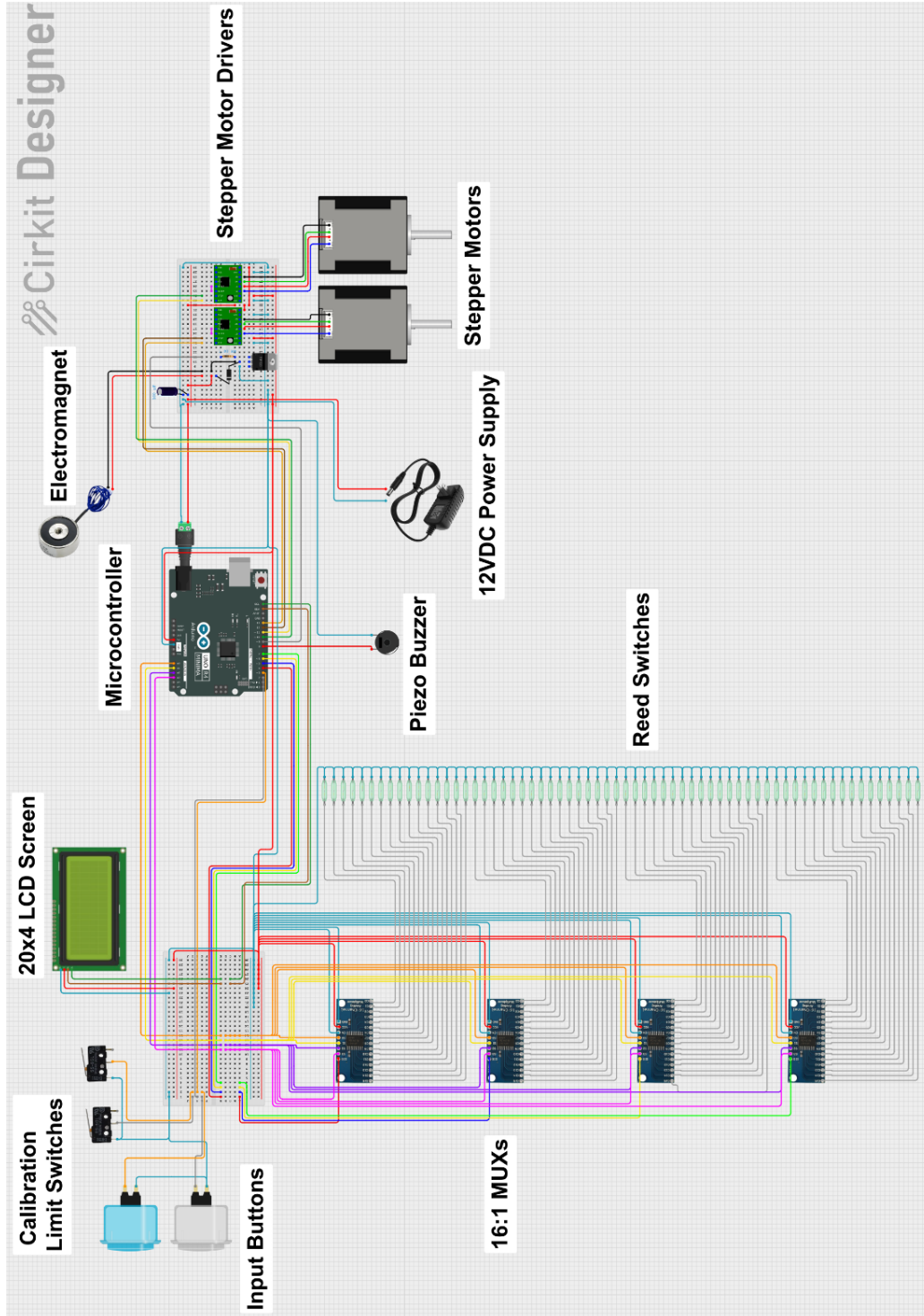


Fig. 2. Circuit Diagram

Device Schematics

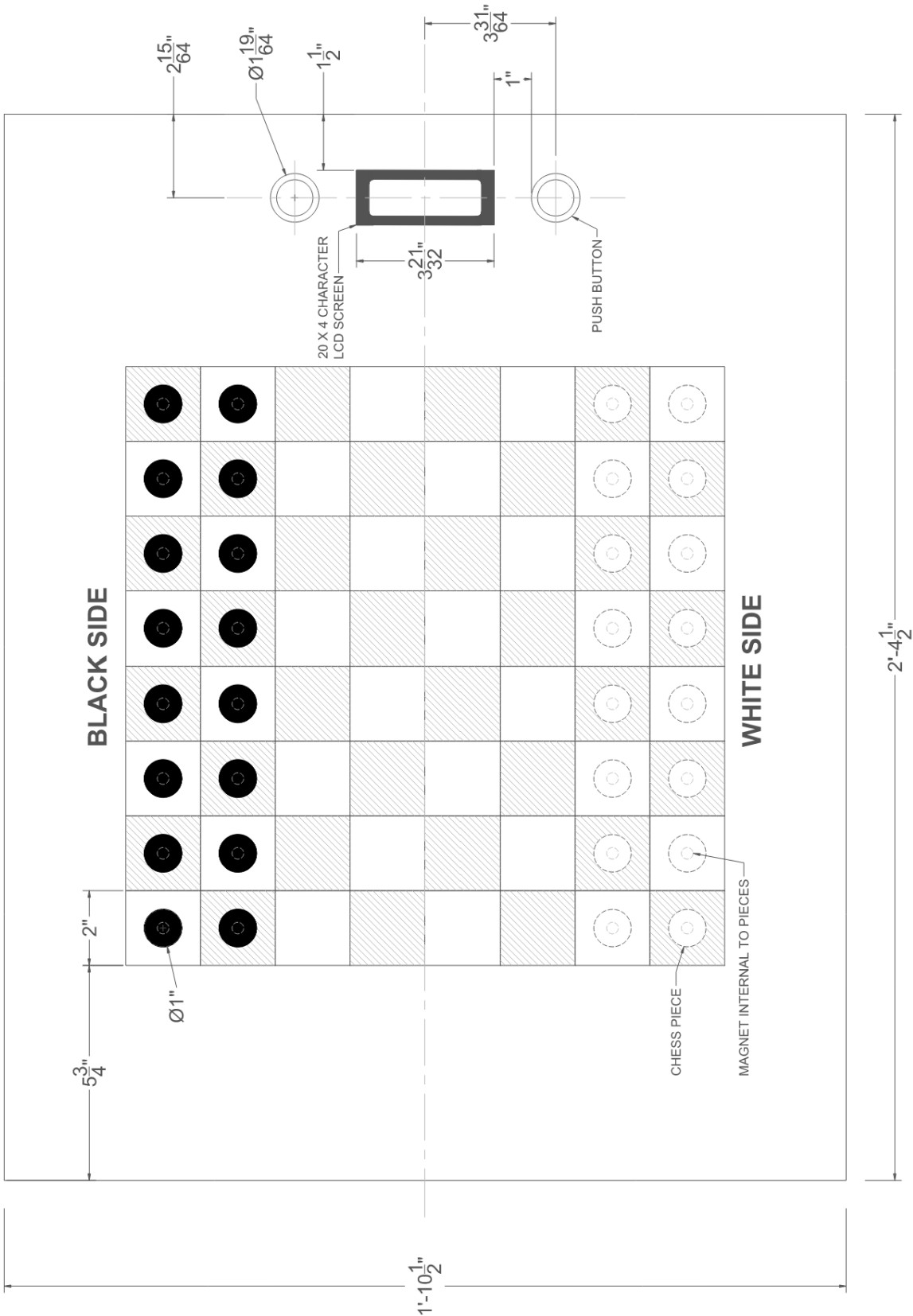


Fig. 3. External View

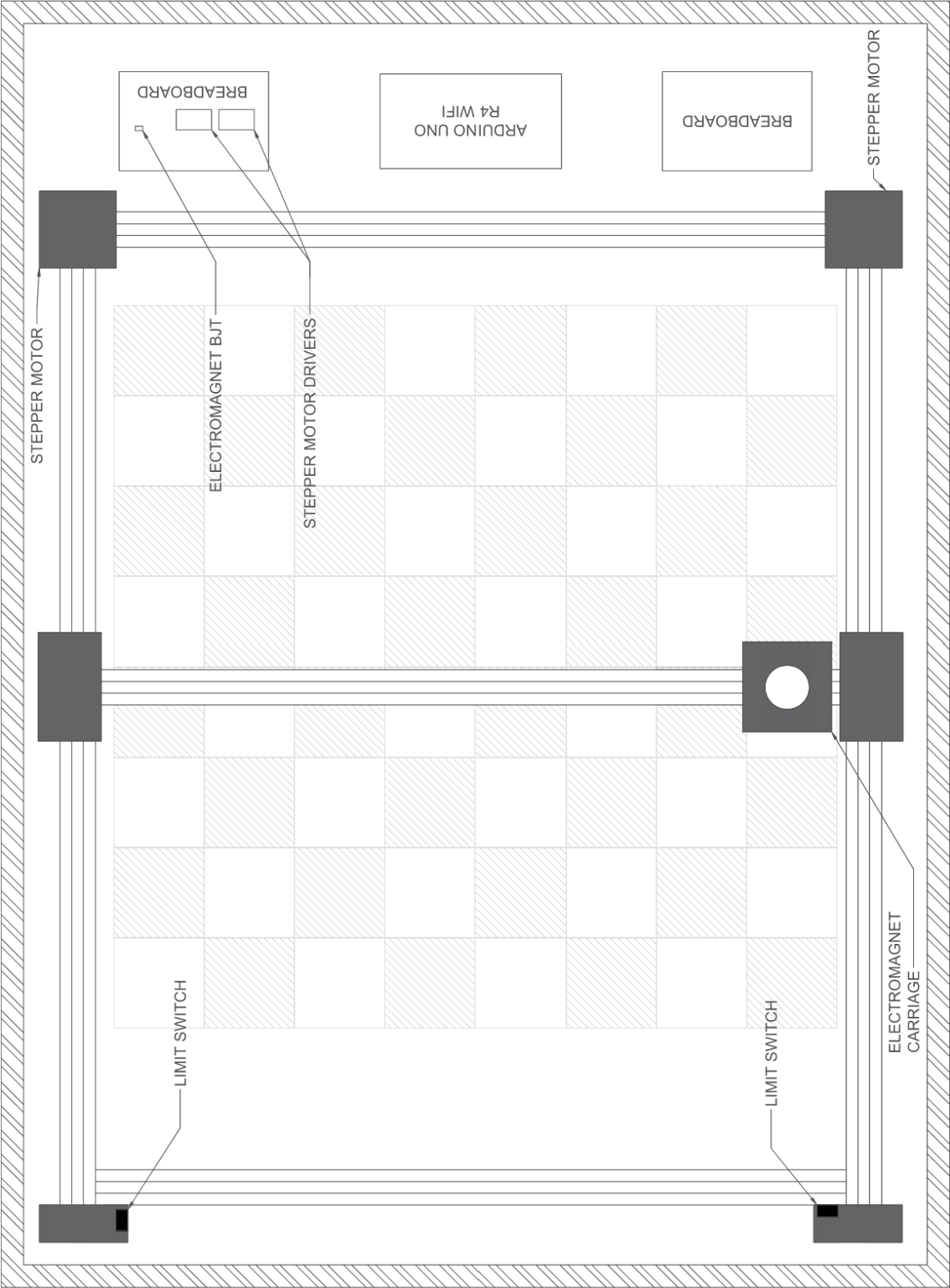


Fig. 4. Internal View

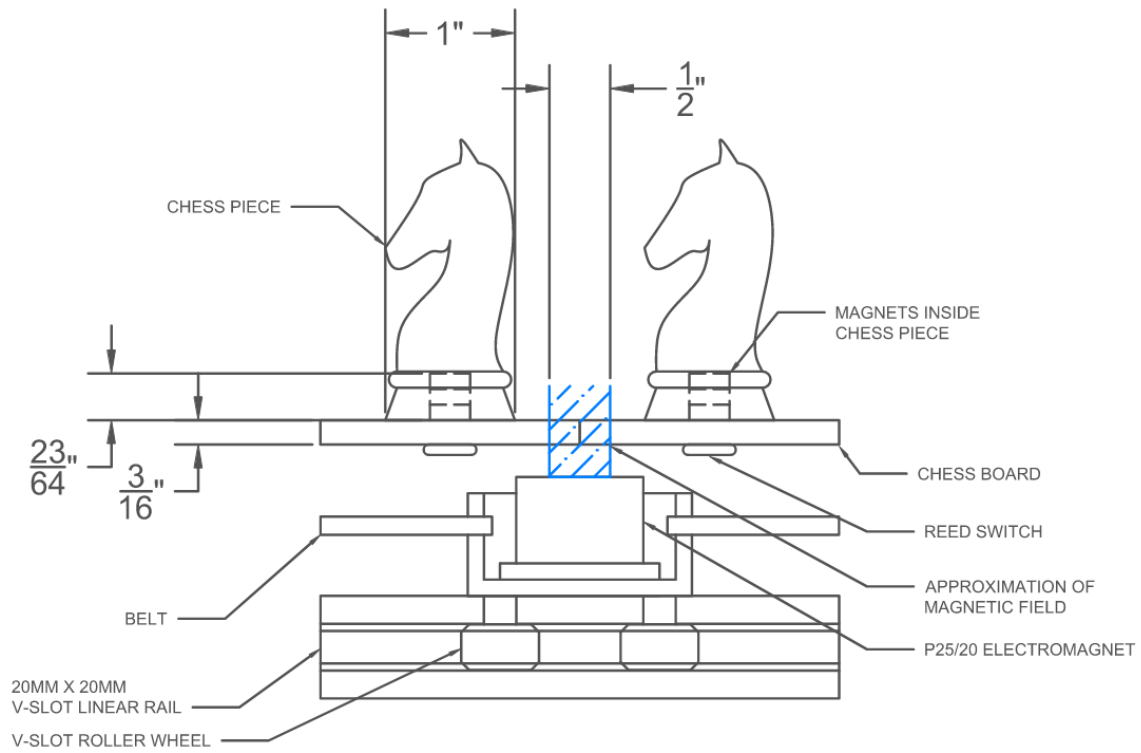


Fig. 5. Enlarged View - Electromagnet Coordination

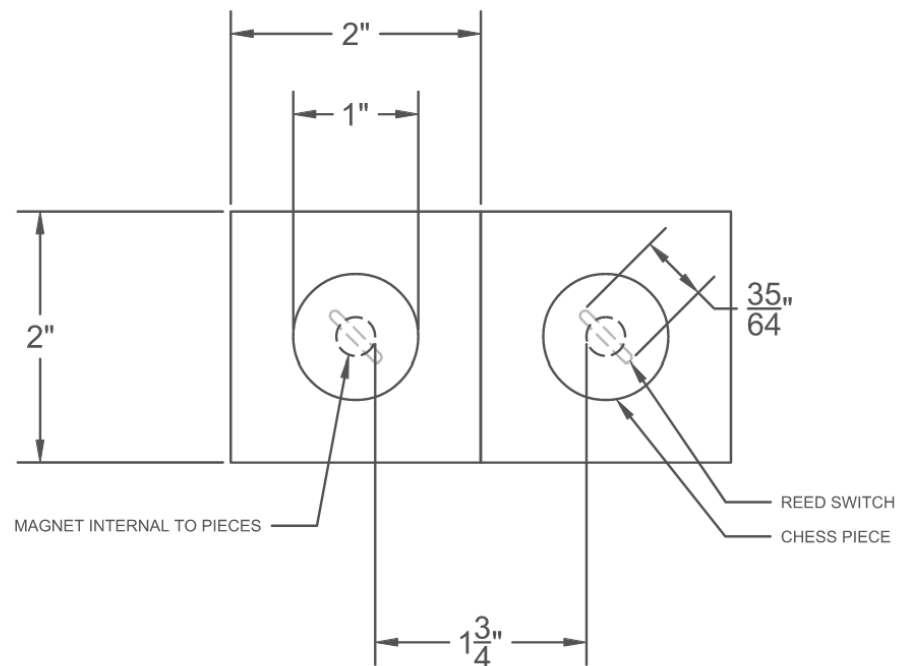


Fig. 6. Enlarged View - Piece Scaling

Bill of Materials

The bill of materials is listed below in Table 1, outlining all materials used to create ChessConnect and the associated information for each. The final material cost of ChessConnect is \$394.22.

Table 1: Bill of Materials

Product Name	Amount	Cost Per Unit	Manufacturer	Standards	Cost
Arduino UNO R4 WiFi	1	\$27.99	Arduino	RoHS Compliant	\$27.99
STEPPERONLINE Nema 17 Stepper Motor	2	\$13.99	STEPPERONLINE	RoHS, NEMA 17, ISO, CE	\$27.98
HiLetgo A4988 Stepstick Stepper Motor Driver Module (5pcs)	1	\$10.19	HiLetgo	N/A	\$10.19
16-Channel Analog/Digital Multiplexer (10pcs)	1	\$9.99	ACEIRMC	IEC 60134	\$9.99
TIP-120 Darlington NPN Transistor (20pcs)	1	\$7.19	BOJACK	N/A	\$7.19
Normally Open Reed Switch (2mm x 14mm) (30pcs)	3	\$8.99	Gebildet	IP54, X-10	\$26.97
Round 12V Electromagnet (100N, 30 x 22mm)	1	\$12.00	TOMSHIELE	IEC 60947-5-1	\$12.00
Piezo Electronic Buzzer (3pcs)	1	\$5.49	WEICHUANGJING	N/A	\$5.49
EG STARTS OEM 30mm Sanwa Push Buttons (12pcs)	1	\$11.99	EG STARTS	N/A	\$11.99
Breadboard Kit (4pcs)	1	\$8.79	REXQualis	EIA RS-310	\$8.79
IIC I2C 20x4 LCD Screen Module (3pcs)	1	\$13.59	Hosyond	RoHS Compliant, I2C, SPI, ISO 9001	\$13.59
8mm x 3mm Neodymium Magnets (50pcs)	3	\$4.79	FINDMAG	ISO 13024, IEC 60404-8-1, EN 71-3	\$14.37
HiLetgo Micro Limit Switch with Roller Lever Arm (10pcs)	1	\$5.99	HiLetgo	IEC 60947, RoHS Compliant	\$5.99
500mm V-Slot 20x20 Linear Rail (10pcs)	1	\$39.99	MECCANIXITY	N/A	\$39.99

GT2 20 teeth Pulley - 5mm bore (5pcs)	1	\$6.99	WINSINN	ISO 5296	\$6.99
Smooth GT2 Idler Pulleys Without Teeth - 5mm bore (8pcs)	1	\$11.99	3Dman	ISO 5296	\$11.99
5m GT2 Timing Belt - Width 6mm	1	\$11.99	MOTOU	ISO 5296, ASTM D2000	\$11.99
Small Pulley V-Shape Wheel 6mm bore (10pcs)	1	\$12.99	befenybay	N/A	\$12.99
Aluminum Spacer 5mm Bore x 10mm OD x 6mm Length (30pcs)	1	\$5.99	uxcell	N/A	\$5.99
Eccentric Spacer 5mm Bore (20pcs)	1	\$8.99	DIAEMI	N/A	\$8.99
.25 kg 3D Printing Filament	1	\$4.88	Likesilk	ISO/ASTM 52900	\$4.88
3/16 in. x 2 ft. x 4 ft. Underlayment Project Panel	1	\$13.34	Home Depot	ANSI/HPVA HP-1, APA PRP-108	\$13.34
1/2 in. x 2 ft. x 4 ft. Radiata Pine Plywood Project Panel	2	\$23.94	Home Depot	ANSI/HPVA HP-1, APA PRP-108	\$47.88
M2 M3 M4 M5 Socket Head Cap Screws Nuts Assortment Kit	1	\$16.89	ZTF	ISO 4762	\$16.89
12V 5A Power Supply	1	\$9.99	Waysse	IEC 62368-1, IEEE C62.41	\$9.99
DC Power Connector 5.5mm x 2.1mm	1	\$3.80	ZHLATECK	EIA-364, IEC 60320	\$3.80
M5 T Slot Nuts for Aluminum Profile (100pcs)	1	\$5.99	HELIFOUNER	N/A	\$5.99
M5 x 20mm 25mm 30mm 35mm 40mm Button Head Socket Cap Screws Bolts	1	\$9.99	Cicidorai	N/A	\$9.99
TOTAL:					\$394.22

User Manual

ChessConnect has very few hardware components that interact directly with the user, making the device extremely simple to use. The main areas of the device that the user should be aware of are the device controls, I/O ports, and procedures for handling pieces. The guidelines for interacting with these areas of the device and other useful information are as follows:

Device Controls – Operating ChessConnect requires only the use of two buttons and an LCD screen which are located on the top panel of the chess board on the same side as the I/O. The buttons are colored black and white to represent the corresponding black and white sides of the chess board. The black and white sides cannot change, or the device will not work properly. During navigation of the home-screen before a chess game has started, pressing the black button will cycle through the various options on a given screen, and pressing the white button will select an option and move the user to the next screen. To go to a previous UI screen, press the black button to cycle through options until “Back” is selected, and press the white button to go back. Once the ChessConnect user enters a game, they will be assigned to either black or white randomly. The button that matches their assigned color is used to simultaneously send their move into the online game and stop their chess clock. Pressing the opponent’s button during a game will bring up an options menu that allows the user to send a game resignation or adjust piece positions. Once in this menu, the opponent’s button will cycle through the various menu options and the user’s button is used to select.

Input/Output Ports – The side of the chess board below the LCD screen and push buttons is where the two I/O ports are located. Looking at the I/O side of the device, the port on the left is a USB type C and the port on the right is a 2.1mm barrel plug that is used to supply 12V power to the device. DO NOT USE ANY POWER SUPPLY THAT IS NOT RATED FOR 12V AND AT

LEAST 5A. Failure to comply with this requirement may result in loss of device functionality and/or destruction of the device. The USB port is connected directly to the microcontroller and can be used for debugging and updating the software which controls the board.

Setting Up/Moving Pieces – Once a game type has been selected, the ChessConnect user will be prompted with a screen to set up the pieces on the board. This screen will show a list of the squares for both the black and white sides that do not have a piece detected on them. At this stage, the user must place all the pieces on their corresponding squares which will cause them to be removed from the screen. Once a side has detected all its pieces, an “all good” message will display. When all pieces on both sides are detected, the request to start a game with the previously selected settings will automatically be sent to Lichess.org. To ensure accurate move detection, the board will print the source and destination square of a piece as the user makes a move. When the user first picks up a piece, they must verify that the screen displays the starting square. Placing the piece back on the starting square will cancel the move that is in progress. When the piece is placed on the destination square, the screen will display the full move and prompt the user to press their button to send the move to the online game. Pressing the opponent’s button will cancel the move and wait for the user to place the piece on the source square before making a new move. If the board is having issues detecting when a piece is picked up or put down, the ChessConnect user should try lightly pushing the piece into the board and moving it around the square. Doing this for every move will increase the consistency of piece detection. The process of taking an opponent’s piece is very similar. The user must first pick up the captured piece, at which point the board will display a “TAKING:” screen with the destination square. Once the captured piece is removed, the capturing piece can be moved into place following the same procedure as previously stated. Similarly, castling requires moving the

king to its destination square first and then moving the rook into position after the screen displays the king's complete move. Note: pieces must be picked up clear of the board and placed on their destination as opposed to sliding. Sliding pieces or hovering could result in incorrect move detection. ChessConnect will automatically promote any pawns to queens. Once a pawn has reached the opponent's side of the board and the move has been sent, the user can replace the pawn with a queen without causing any issues.

Acceptance Testing Procedure and Criteria

ChessConnect was deemed functional after a series of comprehensive tests performed by both the project group and external testers. The primary functionality, quick and accurate piece movement, was tested extensively.

Initially, isolated testing on the internal gantry was done to verify that it could move pieces from within the board. This testing revealed various issues regarding electromagnet strength, consistency of piece movement, and speed of piece movement. These issues resulted in the project teaming making several changes throughout the construction of the device. A stronger electromagnet was bought to overcome the air gap between the board surface and the electromagnet itself as well as maintain the ability to hold onto the piece at higher speeds. Additionally, a new piece magnet solution was established after testing various implementations to determine a balance between piece detection and movement consistency. Finally, a new power supply was ordered as a power limit was encountered while attempting to meet the threshold value for automatic piece move time.

After these solutions were implemented, final testing was conducted to verify that the engineering specifications were met and that all piece movements were consistent and accurate.

During final testing, full games were played on the board against opponents who were unfamiliar with the device. This allowed the project group to observe the simplicity of the device and detect any unforeseen issues. The engineering specification for piece detection errors was monitored during these games and ended up being slightly above our threshold value. As a result, controls were implemented to manually correct any detection errors. Furthermore, testing revealed that repeated diagonal movements cause a reduction in movement accuracy, which caused the team to implement periodic recalibration. As a software test, the project group tracked the internal board state via the serial monitor as games were played. This allowed the group to observe changes as moves were made and detect if the board state had any inconsistencies. Many issues were found during these tests and were resolved to ensure that gameplay was seamless.

After all issues were resolved, outside testers continued playing games using the device. These tests confirmed that the system was easy to use and provided seamless gameplay. The comprehensive testing process, including internal and external evaluation, verified that the system met its functional requirements and was ready for full use.

Software Appendices

Program Description

The program used to control ChessConnect was written in Arduino C++ and runs on the Arduino UNO R4 WiFi. The program manages the operation of the internal gantry and electromagnet for piece movement, scans reed switches for piece detection, and processes user input from the push buttons. This functionality is integrated with game logic and online communication through Lichess.org. Initially, a user can select their gameplay settings and start a game using the push buttons. During a game, responses received from the Lichess server are interpreted and relayed to the user via the LCD display to ensure synchronized gameplay between the board and online opponents. Overall, the software successfully maintains consistent game state tracking, handles movement control, and facilitates user interaction throughout chess games.

Software Flowchart

ChessConnect has three main routines used to achieve seamless gameplay. These routines are game selection, opponent's move, and user's move. Software flowcharts describing these routines, and how they interact, are shown below. Figure 7 outlines game selection, Figure 8 outlines the opponent's move, and Figure 9 outlines the user's move.

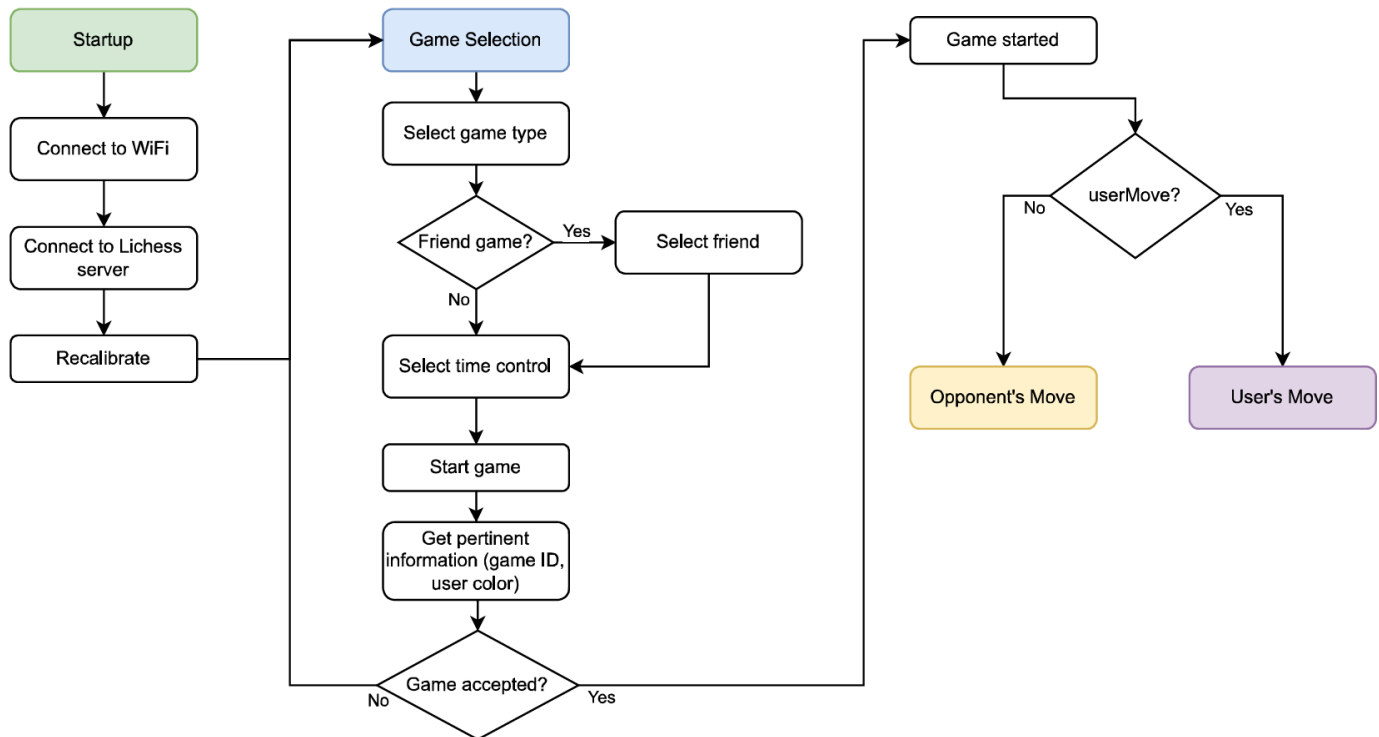


Fig. 7. Game Selection Flowchart

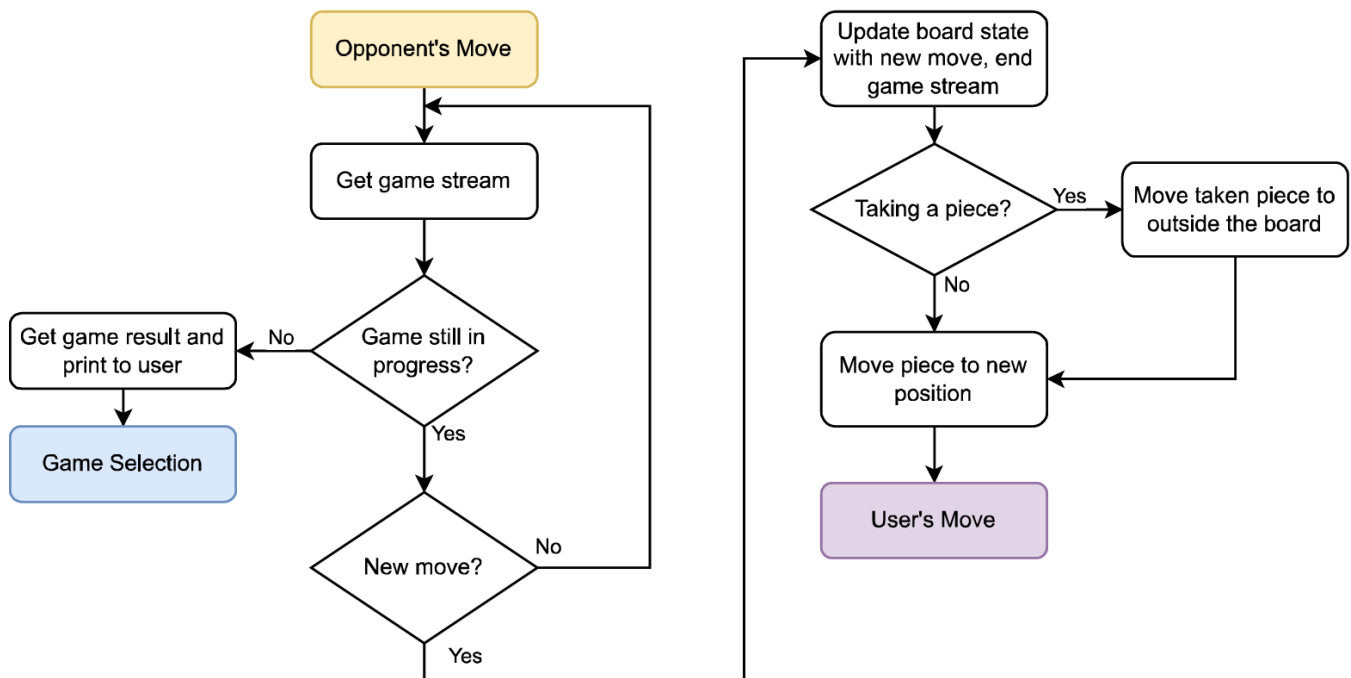


Fig. 8. Opponent's Move Flowchart

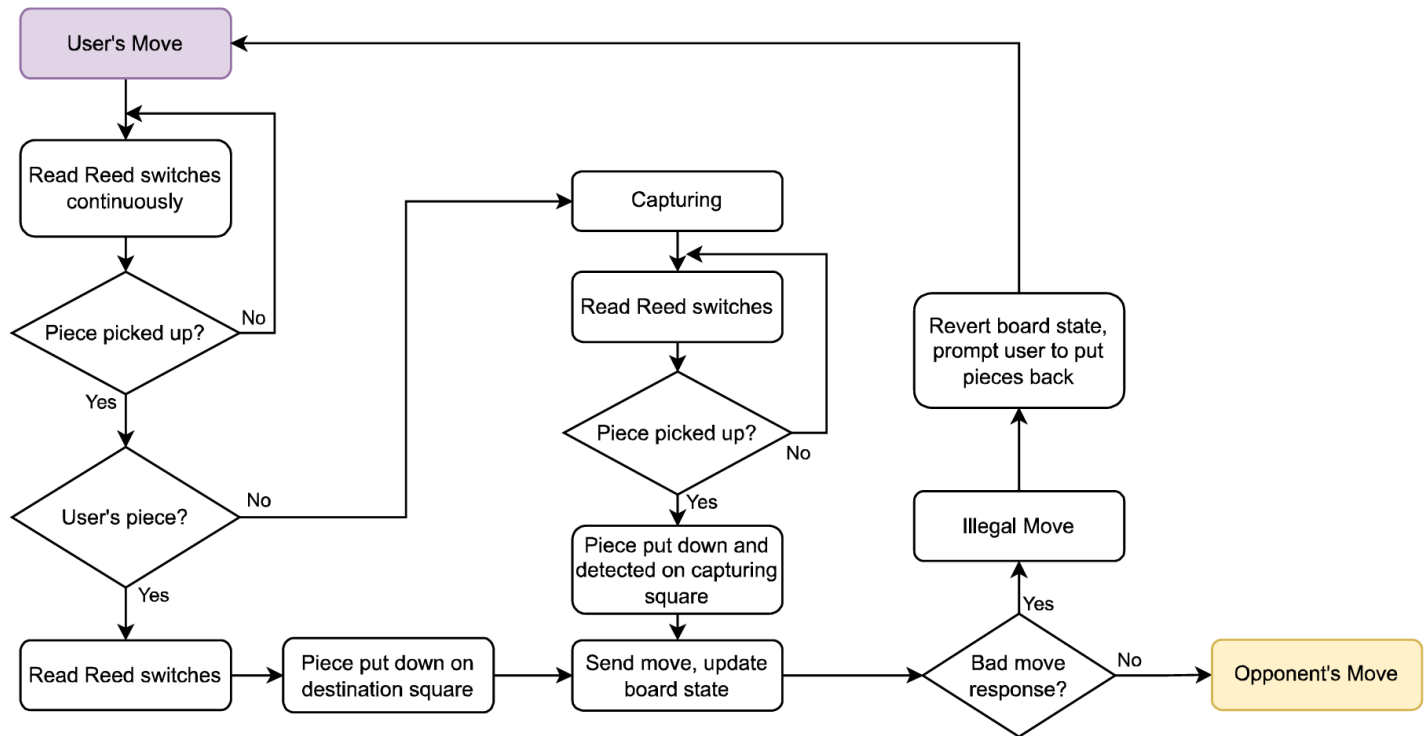


Fig. 9. User's Move Flowchart

Source Code

The complete source code for ChessConnect can be found in the project binder under Tab #13 – Source Code.